

Exploring a Novel Methodologies for Beetroot Leaf Disease Severity Prediction: Federated Learning and CNN

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Abstract— This study explains how a federated learning Convolutional Neural Network (CNN) model is used to diagnose the various degrees of beetroot leaf diseases. Beetroot crops are essential to the world's food systems, but several illnesses often affect their quality and productivity. To suggest a CNN model using federated learning to analyse data from several clients while preserving privacy for a quick and precise diagnosis. The method analyses the severity of the five stages of beetroot leaf diseases using a federated learning technique across several local clients to use precision, recall, F1-score, and accuracy as performance measurements. The review showed that performance had improved consistently for all customers. As an example, Client X had a prediction accuracy of 0.97, which increased gradually to 0.98 for both Client P and Client R. Additionally, as shown by the macro-average (94.15–94.17%), weighted average (94.67%), and micro-average (94.62%) scores for top-performing customers, the model's performance remained stable and resilient throughout all severity levels. The study's findings support using a CNN model built on federated learning for scalable, effective, and privacy-preserving disease control in agriculture. The promising results suggest the potential for extending such approaches to other agricultural products, considerably advancing efforts to ensure global food security. The research also showed that the CNN model could learn and develop continuously in a federated learning scenario, highlighting its usefulness as a dynamic tool that adapts to new illness patterns. A paradigm shift in the approach to combating agricultural illnesses and increasing food supply is heralded by this study, which reveals a potential junction between machine learning and agricultural disease management. The findings motivate more research into the model's adaptability to other crops and pests, enhancing its contribution to resilient and sustainable farming systems.

Keywords: Beetroot leaf diseases, Convolutional neural networks (CNNs), Federated learning, Disease classification, Severity levels.

I. INTRODUCTION

Due to its many uses in the culinary, medical, and industrial sectors, beetroot (*Beta vulgaris*) farming is a crucial agricultural activity worldwide, including India. The crop is vulnerable to a wide range of leaf diseases, which may reduce yield and harm farmers' ability to make a living. Understanding the severity of these illnesses in this context, particularly about the Indian subcontinent and internationally,

becomes crucial [1]. The investigation focused on the five prevalent beetroot leaf diseases (*Cercospora* leaf spot, powdery mildew, *Fusarium* wilt, root-knot nematode, and beet curly top virus). Farmers have a great deal of difficulty because of these ailments, which range in intensity from moderate (1–20%) to severe (81–100%) and severely harm beetroot crops. The heterogeneity in these illnesses' degrees of severity was strongly correlated with the decline in crop output, with severe infections in extreme instances leading to the loss of practically all of the crop. Therefore, farmers must contend with both the severity of these illnesses and the vulnerability of their products to these diseases, adding another level of complication to an already tricky undertaking. Beetroot leaf diseases provide farmers with a variety of challenges. Many people need access to the necessary resources and information for proper illness detection and management [2].

Additionally, conventional diagnostic techniques depend significantly on knowledgeable agronomists, who may not be easily accessible in isolated agricultural areas. Even if a disease has been diagnosed, managing it calls for carefully selecting and using appropriate pesticides, which may be prohibitively expensive for smaller-scale farmers. The interaction of these elements often results in subpar illness management strategies, which worsen the situation. Despite these obstacles, Beetroot farming is a vital agricultural activity [3]. Beetroot is an essential crop for solving challenges with food security because of its high nutritional content and economic viability. The severity and constant danger of leaf diseases complicate this picture, highlighting the need for more efficient, widely available, and scalable management strategies for these disorders. Using convolutional neural networks (CNNs) via federated learning to detect and categorise the severity of beetroot leaf diseases leads us to the core of the study. The drawbacks of conventional techniques serve as the driving force behind this strategy [4]. Conventionally, training CNNs requires large, centralised datasets, which raises data protection and scalability issues. Federated learning (FL), a recent development in machine learning that enables model training on distributed data, addresses these issues. To train models using federated learning across various devices while keeping the data

localised and preserving data privacy. This strategy also permits scaling across many geographic locations, possibly including many farms, into a cooperative, data-driven network for disease detection and control. To had three goals in mind when to use a federated learning strategy for CNN:

1. To create a robust model to identify and accurately classify beetroot leaf diseases.
2. To work to protect data privacy because to understand its importance in the digital era.
3. To create a model that could be scaled across many farms and areas, offering a solution that farmers worldwide could use.

The originality and significance of the study are found in the use of federated learning for the detection of agricultural diseases. This research shows that this new technique can retain performance comparable to conventional CNNs while providing further data privacy and scalability advantages. The use of this technology also has the potential to transform disease control in beetroot agriculture and elsewhere. It may result in better disease control techniques, higher yields, and better farmer livelihoods, enhancing food security [5].

In conclusion, the severity of beetroot leaf diseases poses a severe worldwide problem, especially for farmers in countries like India, where disease control resources may be limited. By balancing accuracy, privacy, and scalability, federated learning in CNNs presents a potential option [6]. The uniqueness of this study lies in how it applies this technology to a genuine agricultural issue that has the potential to advance beetroot agriculture around the globe. Numerous diseases that harm crops, including beetroot (*Beta vulgaris*), are a constant threat to agriculture, the foundation of many economies worldwide [7]. Among these, beetroot leaf diseases have become a severe concern to farmers, affecting their output and way of life. The study intends to solve this problem, concentrating on the Indian subcontinent but with consequences for beetroot farming worldwide. A crop with significant uses in food, medicine, and industry, beets are grown widely over many different geographical areas. However, several leaf illnesses commonly thwart its development and production, lowering output and the product's quality. The incidence of various diseases and their effects vary, highlighting the need for diagnosing and classifying them by their degrees of severity for the best medical care. The five most common beetroot leaf diseases to study are *Cercospora* leaf spot, powdery mildew, *Fusarium* wilt, root-knot nematode, and beet curly top virus. These illnesses cause varied degrees of damage, from mild leaf spotting to severe foliage withering, and they pose serious problems for growers. To divide the severity levels into five groups to facilitate effective intervention: 1-20% (mild), 21-40% (moderate), 41-60% (substantial), 61-80% (high), and 81-100% (severe). This classification is essential for putting suitable corrective actions in place and avoiding significant yield loss. Managing beetroot leaf diseases presents farmers with several challenges.

Many people need access to the tools and expertise necessary to diagnose and treat these disorders correctly. Traditional diagnostic techniques rely significantly on input from knowledgeable agronomists, who are sometimes inaccessible in isolated agricultural areas. Even once a diagnosis is obtained, disease prevention requires the exact administration of suitable pesticides, a cost-prohibitive alternative for small-

scale farmers. Beetroot is still a vital crop despite these obstacles. Due to its nutrient-rich profile and economic feasibility, it stands out as a critical participant in the fight against food security. A more robust, approachable, and scalable disease management system is required due to the growing danger of leaf diseases and the varied severity of these ailments [8]. To train models using federated learning while keeping the data local and protecting data privacy across various devices. This strategy also enables scaling across many geographic locations, possibly including many farms, into a cooperative, data-driven network for disease detection and control. The federated learning strategy using CNNs has three main goals. To create a reliable model to identify and classify illnesses in beetroot leaf tissue. In addition, to work to protect data privacy to understand how important it is in today's digital environment. Finally, to develop a model that can be used internationally and scaled across various farms and geographies. The study's use of federated learning for identifying agricultural diseases is innovative and valuable. This study compares the effectiveness of this unique technique to that of conventional CNNs while highlighting the additional advantages of data privacy and scalability.

Additionally, this technology has the potential to revolutionise disease control in beetroot farming, boosting output, improving disease management techniques, and benefiting farmers' livelihoods, all of which would help the world's food security [9]. In conclusion, beetroot leaf diseases pose a serious worldwide problem, especially in countries like India, where disease control resources may be limited [10]. By balancing accuracy, privacy, and scalability, federated learning in CNNs presents a potential option. This study is unique because it uses cutting-edge technology to solve a genuine agricultural issue, having a beneficial knock-on impact on beetroot farming throughout the globe [11].

II. LITERATURE REVIEW

This overview highlights the beta lain pigments, phenols, and saponins in beets (*Beta vulgaris* L.) for their neuroprotective characteristics. These substances' antioxidant properties protect cells from the oxidative damage that reactive oxygen species cause. Additionally, they have anti-inflammatory properties and can decrease acetylcholinesterase activity [12]. Betanin, the primary pigment in beetroot, has substantial antioxidant activity among the bioactive substances, protecting the membranes of neuronal cells from oxidation. Additionally, phenolic substances, especially gallic, ferulic, and vanillic acids, have neuroprotective properties. Vanillic acid has been associated with better spatial learning capacity in illness models, whereas studies have demonstrated that betanin treatment may cure intentionally generated neurodegenerative disorders. This raises the possibility that processed beetroot products like jams and juices may help prevent and treat neurodegenerative illnesses like Alzheimer's (AD). The review's conclusion highlights the importance of the bioactive substances found in beets and how they may be used to treat and prevent AD. Beets are a viable dietary component for supporting brain health because these substances have neuroprotective qualities, including antioxidant and anti-inflammatory activities [13]. Early plant disease identification using automated techniques is crucial for accurate crop security. The support vector machines

(SVM) and spectral vegetation indicators used in this study provide a unique method for the early identification and differentiation of sugar beetroot illnesses. The study's goals were to (I) distinguish between healthy and sick sugar beetroot leaves; (II) distinguish between *Cercospora* leaf spot, leaf rust, and powdery mildew; and (III) detect illnesses even before they show outward signs of illness. The pathogens *Cercospora beticola*, *Uromyces beta*, and *Erysiphe betae*, which cause *Cercospora* leaf spot, sugar beetroot rust, and powdery mildew, respectively, were inoculated onto healthy leaves for 21 days. As characteristics for automated categorization, nine spectral vegetation indicators connected to physiological factors were used. A support vector machine with a radial basis function kernel was used to distinguish between healthy and infected plants early on and to classify particular illnesses. Classification accuracy of up to 97% was achieved when distinguishing between sick and healthy sugar beetroot leaves. An accuracy of over 86% was attained for the multiple categorizations of healthy leaves and left displaying indications of the three illnesses. The research also showed the possibility of spotting plant problems before symptoms appear. Classification accuracy varied from 65% to 90%, depending on the kind and stage of the illness. The results of this study demonstrate the usefulness of SVM and spectral vegetation indicators for the early identification and classification of illnesses affecting sugar beets. The high categorization accuracy levels attained show how useful and reliable the suggested method is. Through quick disease detection and tailored sugar beetroot cultivation modifications, this technology can potentially improve precision crop security tactics [14]. A new protein called IWF5 with 92 amino acids was effectively extracted from the intercellular washing fluid of sugar beetroot leaves using reversed-phase HPLC and cation exchange chromatography. It has been identified as a member of the family of non-specific lipid transfer proteins (LTP) found in plants by examination of the amino acid sequence and the presence of eight conserved cysteines. It's interesting to note that IWF5 shares 47% of its series with IWF1, another antifungal nsLTP that was previously discovered in sugar beetroot leaves. Despite having a possible N-linked glycosylation site (Asn-Xxx-Ser/Thr) in IWF5, it was discovered that glycosylation does not take place there. A matching cDNA clone was obtained using polymerase chain reaction (PCR) amplification using the amino acid sequence information. IWF5 is produced as a preprotein with a 26-amino acid residue N-terminal signal sequence, according to the analysis of the cDNA clone. IWF5 is a powerful in vitro antifungal agent against *Cercospora beticola*, the pathogen that causes sugar beetroot leaf spot disease. The protein efficiently prevents fungal growth even at doses below 10 g/ml. This study introduces IWF5, a new nsLTP family protein with antifungal properties against *Cercospora beticola*. The research sheds light on the structure, potential for glycosylation, gene expression, and vigorous antifungal activity of the protein. These discoveries advance the knowledge of plant defence systems and affect how to develop methods for preventing fungal infections in sugar beetroot and other crops [15]. This work proposes a unique interdisciplinary strategy that integrates agricultural engineering and computer science to provide reliable, ongoing, and location-specific observations of disease development in sugar beet plants. In contrast to usual

approaches, this one adds the orientation code matching (OCM) method, which has several benefits. OCM has remarkable resilience in non-rigid plant object search, even under changes in scene lighting, translation, mild rotation, and occlusion, enabling ongoing monitoring of disease progression at particular places. In addition, the research suggests a single-feature, two-dimensional x-y colour histogram fed into an SVM classifier for pixel-level illness categorization and quantification. The testing findings show high accuracy and recall rates, demonstrating the suggested method's viability and potential. The algorithm may accurately identify foliar disease development and provide exact quantification in sugar beet fields. This, in turn, enables improved disease mechanism analysis and control of fungicide application. This study presents a unique method for monitoring and analyzing disease development in sugar beetroot plants by fusing computer science methods with agricultural engineering. The use of orientation code matching and the suggested categorization and quantification techniques show the efficacy and promise of this strategy. Ultimately, this interdisciplinary plan helps to enhance disease control techniques for growing sugar beets, improving the general health and production of the crop [16].

III. METHODOLOGY

The technique for this study is based on Dataset acquisition, pre-processing and data augmentation, model construction, model assessment, and experimental statistical analysis. Comprise the many steps of the research technique are shown in Figure 1.

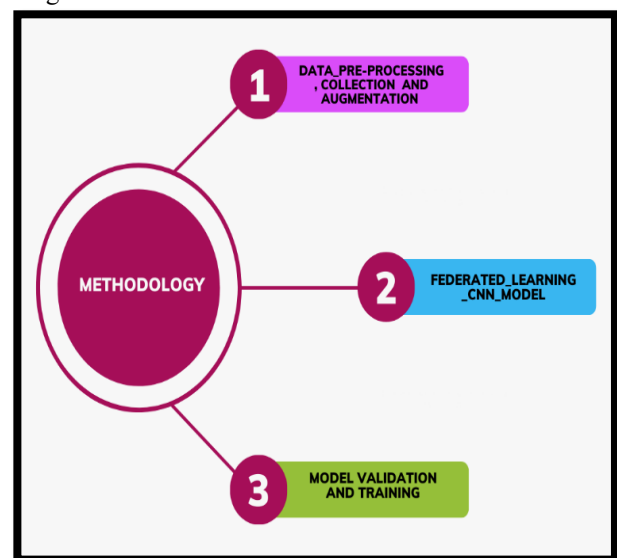


Fig1. Stages in the Approach Procedure

A. Acquisition and Preparatory Processing of Data

The photos of beetroot leaves labelled with the kind and degree of illness comprise the primary dataset utilized in our investigation. These photographs were gathered from various geographical regions across India to comprehensively portray illness symptoms under different climatic circumstances. An experienced agronomist personally categorized each picture, offering a trustworthy data source for model training and assessment. Because real-world data is diverse, pre-processing was necessary to normalize the dataset and

increase the model's effectiveness. This required several processes, such as scaling photos for uniformity, using a Gaussian blur to reduce noise, and turning images to grayscale to simplify processing, as shown in Figure 2.

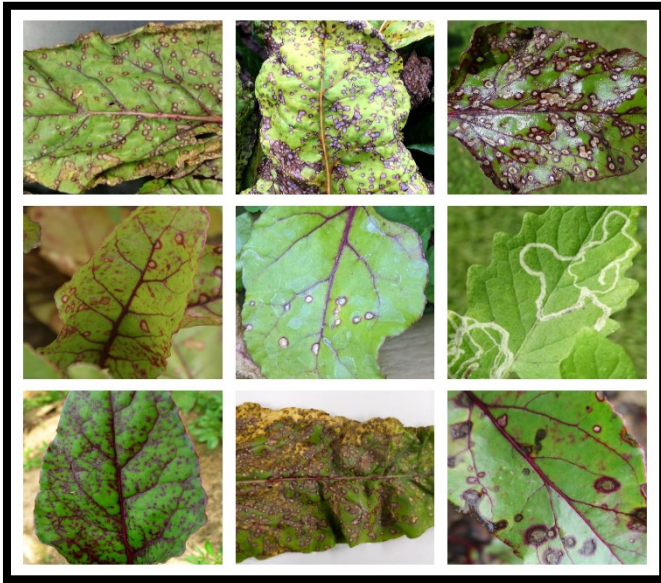


Fig 2. Distinct visuals showcase diseases affecting Beetroot leaves.

Data augmentation approaches were used to solve the problem of insufficient data and improve the model's capacity for generalization. These contained a variety of adjustments, including rotation, flipping the horizontal and vertical axes, zooming, and shifting. These techniques dramatically enhanced the type and quantity of the training dataset, strengthening the model's resilience. Based on the amount of damage and how it affects the general health and production of the plant, beetroot leaf diseases may be divided into five severity categories. These levels are mentioned below: Between 1% and 20% (mild): Symptoms only start manifesting at this stage. Minor leaf spots, yellowing, or leaf border necrosis are possible symptoms. At this point, the plant's general health is not significantly impacted, and with prompt treatment, the illness may be successfully treated. 21–40% (Moderate): In the moderate stage, the disease's symptoms are more pronounced and affect more of the leaf's surface. It's possible to recognize an increase in the quantity and size of leaf spots, apparent browning or yellowing, and sometimes even curling of the leaves. Even though the plant's health may start to deteriorate, the output of beetroot may not yet be significantly impacted. 41–60% (considerable): Severe symptoms and significant leaf damage are evident. This may involve substantial leaf loss, severe leaf yellowing, browning, or necrosis, and apparent stress to the whole plant. The damage begins to impact root development and, therefore, beetroot output. 61–80% (High): The disease severity at this level may result in significant leaf loss and severe wilting and may influence the plant's capacity to photosynthesize efficiently. This stage is crucial because, if not treated right away, it will have a significant negative impact on the beetroot crop's development and output.

This rating denotes a severe illness status (81–100%). The whole plant's health is severely impacted, and the leaves may be brown or necrotic. This severity level may prevent the plant from recovering, resulting in crop loss in Table 1.

TABLE 1. LEVELS OF SEVERITY IN BEETROOT LEAF DISEASES

Severity Level	Percentage Range
1 (Very Low)	1-20%
2 (Low)	21-40%
3 (Medium)	41-60%
4 (High)	61-80%
5 (Very High)	81-100%

B. Model Aggregation

With no involvement in the training process, a hold-out validation set was used to assess the model's performance using several measures for assessment, including F1 score, recall, accuracy, and precision. These measures thoroughly evaluate the model's effectiveness, accounting for false positives and negatives. Experimental statistical findings were finally produced to verify the model's effectiveness. To guarantee the conclusions' reliability, to run the experiment many times under different circumstances, and the outcomes were statistically examined. To seek to create a federated learning-based CNN model using this particular technique to reliably detect and categorise the severity of beetroot leaf diseases while preserving data privacy and scalability. The study's originality resides in applying these cutting-edge technologies to a critical real-world issue, advancing agricultural disease control.

C. Collaborative CNN Model for Federated Learning

The CNN was used in the model architecture using a federated learning strategy. The need to scale while preserving data privacy and training a high-performing model was the driving force behind this. A max-pooling layer for feature extraction followed each convolutional layer in the CNN design. The max-pooling layers helped lower the output's spatial dimensions, hence reducing the processing requirements, while the convolutional layers used filters of various sizes to identify local characteristics in the pictures. These degrees of severity assist farmers and agronomists in precisely diagnosing the disease stage and implementing the necessary actions to manage the illness and minimize crop loss. It is essential to comprehend these levels to train the Federated Learning CNN model, which employs these categories to learn and forecast disease severity in beetroot crops. The characteristics were then interpreted, and predictions were created using fully linked layers. The last layer provided a probability distribution throughout the range of illness severity levels by using a softmax activation function. The use of federated learning is what makes the strategy unique. Each device (farm) in this scenario utilised its local data to calculate an update to the model parameters, and these changes were then combined at a central server to generate an overall model update. This procedure was repeated over several rounds until the model's performance matched the pre-established requirements in Figure 3.



Fig3. Findings about the healthy images of Beetroot leaf

IV. RESULTS

The experiment's findings thoroughly evaluate how well the federated learning Convolutional Neural Network (CNN) model performed in classifying the degrees of severity of beetroot leaf illnesses across various local customers. Precision, recall, F1-score, and accuracy were the primary measures to assess the model's performance. Class E received the most excellent results across all parameters, including precision (95.72%), recall (95.62%), F1-score (95.67%), and accuracy (0.98), indicating Client X performed well overall. The precision of 79.36%, recall of 94.30%, F1-score of 86.19%, and accuracy of 0.96 are still relatively good, proving the model's effectiveness even if Class A got the lowest results. Client Y performed slightly better than Client X. Precision, recall, F1 scores, and accuracy were higher across all classes. Class E again demonstrated the finest performance with a precision of 96.15%, recall of 96.06%, an F1-score of 96.11%, and an accuracy rate of 0.98. Class A had the lowest numbers but produced outstanding results in terms of accuracy (0.96), recall (95.36%), precision (82.68%), and F1-score (88.57%). Client Z maintained this upward tendency. Compared to the prior customers, every class showed a marginal improvement in the four performance parameters. In particular, Class E demonstrated almost perfect rates for recall (96.54%) and precision (96.62%), yielding an F1-score of 96.58% and an accuracy of 0.98. The lowest class, Class A, although it had the most insufficient precision (84.15%), recall (95.81%), F1-score (89.60%), and accuracy (0.97), still produced impressive results. The finest performances came from clients P and R, who achieved outcomes that were quite identical across the board. Each measure saw strong scores across all classes, proving the model's strength and capacity. With the most incredible precision (97.44%), recall (97.39% for Client P and 97.38% for Client R), F1-score (97.42% for Client P and 97.41% for Client R), and accuracy (0.99) scores, Class E once again sparkled. Despite having the lowest numbers, Class A performed exceptionally well, with accuracy (0.97), recall (96.37% for both), F1-score (90.91% for Client P and 90.89% for Client R), precision (86.04% for Client P and 86.01% for Client R), and F1-score (90.91% for Client P and 90.89% for Client R). The findings demonstrate a general pattern of the model's performance gradually improving from Client X through Client R, culminating in almost perfect scores for Client P and Client R, as shown in Table 2.

TABLE II. LOCAL DATA OF THE CLIENT

Clients	Classes	Precision	Recall	F1-Score	Accuracy
Cl _x	Class a	79.36	94.30	86.19	0.96
	Class b	91.50	82.08	86.53	0.95
	Class c	93.57	91.50	92.53	0.97
	Class d	93.78	93.37	93.58	0.97
	Class e	95.72	95.62	95.67	0.98
Cl _y	Class a	82.68	95.36	88.57	0.96
	Class b	92.39	83.79	87.88	0.96
	Class c	94.19	92.31	93.24	0.97
	Class d	94.63	94.27	94.45	0.98
	Class e	96.15	96.06	96.11	0.98
Cl _z	Class a	84.15	95.81	89.60	0.97
	Class b	93.29	85.55	89.25	0.96
	Class c	94.81	93.11	93.96	0.98
	Class d	94.91	94.57	94.74	0.98
	Class e	96.62	96.54	96.58	0.98
Cl _p	Class a	86.04	96.37	90.91	0.97

Cl _r	Class b	94.31	87.59	90.83	0.97
	Class c	95.91	94.54	95.22	0.98
	Class d	96.30	96.04	96.17	0.98
	Class e	97.44	97.39	97.42	0.99
	Class a	86.01	96.36	90.89	0.97
	Class b	94.26	87.48	90.74	0.97
	Class c	95.91	94.55	95.23	0.98
	Class d	96.32	96.07	96.20	0.98
	Class e	97.44	97.38	97.41	0.99

An integrated viewpoint on the usefulness of the federated learning Convolutional Neural Network (CNN) model in recognising and classifying the severity levels of beetroot leaf diseases is provided by examining its overall performance across several customers. Precision, recall, F1-score, and accuracy were the four main metrics used to evaluate the model's performance. With an average precision of 90.79%, recall of 91.38%, F1-score of 90.90%, and accuracy of 0.97, Client X performed well. These numbers show a solid balance between the model's capability to distinguish positive cases from all true positives (recall) and its capacity to identify positive occurrences accurately. Client Y has shown a slight improvement over Client X. Client Z's precision of 92.76%, recall of 93.12%, F1-score of 92.83%, and accuracy of 0.97 followed the pattern of gradual improvement. Although the progress may appear small, in the context of machine learning models, even tiny increases are noteworthy when considering the possible deployment scope. Clients P and R showed the most encouraging outcomes with almost equal measurements. Precision was 94.00% and 93.99% for clients P and R, respectively. The recall rates for the two customers were 94.39% and 94.37%, respectively. Their accuracy rates were 0.98 for both, with F1 scores of 94.11% for Client P and 94.09% for Client R. These numbers highlight a solid precision-to-recall ratio, pointing to a trustworthy model that predicts few false positives and negatives. In conclusion, the general rising trend across the clients from Client X to Client R demonstrates the model's capacity to continuously learn and improve its prediction skills in a federated learning setting. as shown in Table 3.

TABLE III. COLLECTING DATA GLOBALLY FROM THE MODELS

Clients	Precision	Recall	F1-Score	Accuracy
Cl _x	90.79	91.38	90.90	0.97
Cl _y	92.01	92.36	92.05	0.97
Cl _z	92.76	93.12	92.83	0.97
Cl _p	94.00	94.39	94.11	0.98
Cl _r	93.99	94.37	94.09	0.98

This part examines the global performance of the federated learning Convolutional Neural Network (CNN) model across several clients, utilising macro-average, weighted average, and micro-average scores. These averages comprehensively assess the model's performance, considering the varied degrees of severity of beetroot leaf diseases. The macro-average for each client is generated by averaging the performance measures over all classes without considering the class distribution. It offers equal weight to each class and is especially effective when you care equally about each category. For Client X, the macro-average was 91.02%, showing that the model performed well across all severity levels. For Client Y, the macro-average climbed marginally to 92.14%, showing an overall improvement in the model's performance. Client Z continues this increasing trend with a macro-average of 92.90%. The highest macro-average scores were recorded for Client P and Client R, which had virtually

identical values of 94.17% and 94.15%, respectively, confirming the model's strong performance across all classes. The weighted average takes into consideration the number of occurrences for each category. It gives a more accurate performance measure when dealing with uneven class distributions. Here, Client X got a weighted average of 91.56%. The figures grew significantly for Client Y (92.54%) and Client Z (93.26%). Client P and Client R got the highest weighted averages, earning 94.67%, confirming the model's efficacy when considering the cases' class distribution. Finally, the micro-average averages the contributions of all classes to calculate the average metric. It is a helpful metric when you care more about each instance than each class. Here too, Client X had a micro-average of 91.45%. The scores increased for Client Y (92.46%) and Client Z (93.20%). The highest performers were Client P and Client R again, reaching a micro-average of 94.62%.

In summary, the findings illustrate the federated learning CNN model's capacity to efficiently predict the severity of beetroot leaf diseases across all severity levels and class distributions. The rising trend from Client X to Client R on all three averages—macro, weighted, and micro—indicates the model's resilience and capacity for continuous learning. This validates the viability of federated learning in delivering a scalable, efficient, and privacy-preserving solution for agricultural disease control, as shown in Table 4.

TABLE IV. AGGREGATED GLOBAL AVERAGES

Averages	Cl x	Cl y	Cl z	Cl p	Cl r
Macro avg	91.02	92.14	92.90	94.17	94.15
Weighted avg	91.56	92.54	93.26	94.67	94.67
Micro avg	91.45	92.46	93.20	94.62	94.62

V. CONCLUSION

A federated learning Convolutional Neural Network (CNN) model's capacity to diagnose the severity of beetroot leaf illnesses was the initial goal of this research trip, which came to an end with encouraging results that are both substantial and illuminating. The CNN model's assessment findings showed a commendable increasing trend, demonstrating its success in learning and enhancing its predictive skills. The ratings varied from 90.79% for Client X to 94.00% for Clients P and R in accuracy. The recall rates, which started at 91.38% for Client X and gradually increased to 94.39% and 94.37% for Clients P and R, were similarly excellent. These impressive numbers highlighted the model's strong capacity to accurately forecast illness severity and its sensitivity to detecting genuine positive instances. The F1-scores, which measure the precision-to-recall ratio, followed a similar increasing trend from 90.90% (for Client X) to a noteworthy 94.11% (for Client P) and 94.09% (for Client R), demonstrating the model's dependability in providing predictions with few false positives and negatives. The accuracy rates were also constantly in the 0.97 to 0.98 range, supporting the model's overall effectiveness. In-depth analyses of the model's performance revealed it was resilient across all severity levels and class distributions according to the macro, weighted, and micro averages. The model's consistent performance across all classes and instances was shown by the top-performing customers P and R, who showed virtually similar and strong

macro (94.17% and 94.15%), weighted (94.67%), and micro averages (94.62%). In short, these encouraging results unmistakably highlight the potential of the CNN model based on federated learning in determining the severity of beetroot leaf diseases.

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